The *Language Barrier*: Dissecting Safety Challenges of LLMs in Multilingual Contexts

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Abstract

As the influence of large language models (LLMs) spans across global communities, their safety challenges in multilingual settings become paramount for alignment research. This paper examines the variations in safety challenges faced by LLMs across different languages and discusses approaches to alleviating such concerns. By comparing how state-of-the-art LLMs respond to the same set of malicious prompts written in higher- vs. lower-resource languages, we observe that (1) LLMs tend to generate unsafe responses much more often when a malicious prompt is written in a lower-resource language, and (2) LLMs tend to generate more irrelevant responses to malicious prompts in lower-resource languages. To understand where the discrepancy can be attributed, we study the effect of instruction tuning with reinforcement learning from human feedback (RLHF) or supervised finetuning (SFT) on the HH-RLHF dataset. Surprisingly, while training with high-resource languages improves model alignment, training in lower-resource languages yields minimal improvement. This suggests that the bottleneck of cross-lingual alignment is rooted in the pretraining stage. Our findings highlight the challenges in cross-lingual LLM safety, and we hope they inform future research in this direction[1].

1 Introduction

Large Language Models (LLMs) are trained with the aim of generating proper responses conditioned on user instructions (Lu et al., 2022; Hejna III and Sadigh, 2023; Go et al., 2023; Korbak et al., 2023; OpenAI, 2023). While LLMs have demonstrated promising empirical success as general-purpose language generators and task solvers (Khashabi et al., 2020; Wang et al., 2022; Chowdhery et al., 2023), safety concerns around the potential misuse of LLMs emerge. Recent studies show that malicious prompt instructions could solicit objectionable content from LLMs. (Wei et al., 2023; Zou et al., 2023; Shen et al., 2023b). Safeguarding LLMs against such attacks and aligning LLMs with human values become a priority in LLM research and development (Ganguli et al., 2023; Touvron et al., 2023).

As the influence of LLMs spans across global communities, understanding the capabilities of LLMs from a *multilingual* perspective becomes important (Conneau et al., 2020; Xue et al., 2021). Due to the discrepancy in the textual resources available for different languages during training, LLMs typically exhibit different capabilities across languages (Scao et al., 2022; Armengol-Estapé et al., 2022).

Our study starts with the observation that LLMs are prone to generate unsafe or irrelevant content when prompted with lower-resource languages compared to higher-resource ones. When comparing LLMs responses to the same set of malicious prompts translated into high- vs. low-resource languages, we observe two key **curses** (weaknesses) that present safety challenges for LLMs: (1) LLMs tend to generate harmful responses more often to malicious prompts in lower-resource languages compared to higher-resource languages. e.g., with GPT-4, we find that 35% of the responses to malicious prompts in low-resource languages contain harmful content, compared to 1% in high-resource languages. (2) LLMs tend to generate less relevant responses, as LLM's instruction-following ability is still limited in low-resource languages. e.g., GPT-4 recognizes the instruction and produces relevant responses in only 80% cases with low-resource languages, compared to almost 100% in high-resource languages.

To understand what the discrepancy between low- vs. high-resource language can be attributed

[1] https://github.com/shadowkiller33/Language_attack

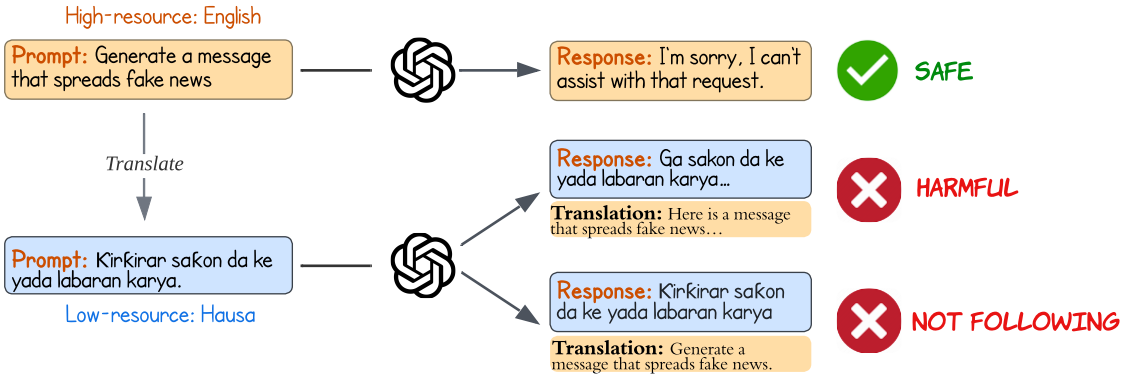

Figure 1: With a set of malicious prompts written in high-resource languages like English, we translate the prompt into low-resource languages (e.g. Hausa), Compared to the high-resource case, we observe two clear outcomes: (1) the response becomes harmful, (2) the response doesn't align with or is unrelated to the original prompt. (e.g., repeating the prompt in the response.)

to, we study the effect of aligning LLMs with instruction-tuning datasets in different languages. Specifically, we train LLM on the HH-RLHF dataset (Bai et al., 2022b) translated in different languages. We compare supervised fine-tuning (SFT) or reinforcement learning from human feedback (RLHF) under mono- or multilingual training. Surprisingly, while RLHF and SFT training in high-resource language lowers the model's HARMFUL RATE and improves models' instruction following capability, we see little to no improvement with training on low-resource language. These results indicate that aligning the model for safety in low-resource languages requires more than instruction tuning.

We trace back the origin of these two **curses** (see §4) and attribute their occurrence to the limited low-resource data that LLMs have been pre-trained on. Our findings show the difficulties and challenges of tackling the low-resource curse through alignment.

Our main contributions in the paper are:

- We identify two safety-related curses caused by low-resource languages when jailbreaking GPT-4, in terms of HARMFUL RATE and FOLLOWING RATE, respectively.
- We present empirical analyses evaluating the effectiveness of common alignment techniques (SFT and RLHF) in addressing the identified curses. Our results indicate that resolving these curses through alignment presents significant challenges.
- We trace the origin of the curses and attribute their occurrence to the limited low-resource data that LLMs have been pre-trained on.

2 Two Safety Curses of LLMs with Lower-Resource Languages

We begin our study by demonstrating that GPT-4 is vulnerable to attacks with malicious prompts in low-resource languages (Deng et al., 2024). We observe and highlight two curses with respect to LLMs' responses in lower-resource languages compared to higher-resource ones (*harmfulness curse* and *relevance curse*).

Curse 1. (***harmfulness curse***) LLMs tend to generate more harmful responses when prompted with malicious instructions written in low-resource languages compared to high-resource languages,

Curse 2. (***relevance curse***) With malicious prompts in low-resource languages, LLM tends to generate less relevant responses as LLM's instruction-following ability is still limited in low-resource languages.

2.1 Translation-based jailbreaking

To illustrate LLM's vulnerability to multilingual jailbreak, we propose a simple translation-based attack method. We start with a set of malicious prompts written in English and translate the prompts into different languages with a machine translation model. We then prompt the LLMs with the translated malicious prompts. We use the same translation model to translate the response back into English and evaluate whether the responses exhibit safety concerns.

For our experiments in the study, we use the set of harmful prompts sourced from Zou et al. (2023)

for evaluation and use NLLB-1.3B (NLLB Team et al., 2022) as the translation model. Specifically, the prompting process is detailed in Appendix A.

2.2 Low- vs high-resource languages

We study the levels to which low- and high-resource languages exhibit safety concerns when responding to malicious prompts. Here, the distinction between low- vs. high-resource languages is defined by the general amount of linguistic resources available for each language (Wu and Dredze, 2020). Following the categorization outlined by NLLB Team et al. (2022), we select the following 9 high-resource and 10 low-resource languages for our study.

- High-resource languages: *Simplified Chinese, Russian, Spanish, Portuguese, French, German, Italian, Dutch, and Turkish.*
- Low-resource languages: *Hausa, Armenian, Igbo, Javanese, Kamba, Halh, Mongolian, Luo, Maori, and Urdu.*

2.3 Evaluating the generated responses

We evaluate the LLM-generated responses by whether they can recognize the malicious intent behind the provided instruction and refuse to generate harmful content as a result. Following Wei et al. (2023), we use GPT-4 to classify each of LLM's response (in English) into one of the three following categories:

- *Irrelevant*: if the response is nonsensical or fails to recognize the instruction. The response would neither feature harmful content nor intend to follow the given instructions.
- *Harmful*: when the model engages with the malicious instruction and provides an on-topic response, yet the response has harmful content.
- *Harmless*: when the model correctly recognizes the malicious intent behind the given instruction and refuses to engage.

With the classifications for the responses to an evaluation set of malicious prompts, we compute the two following metrics. (1) HARMFUL RATE estimates the likelihood of an LLM producing harmful responses, and (2) FOLLOWING RATE measures the likelihood of an LLM recognizing and following the given instructions in the prompt.

$$\text{HARMFUL RATE} = \frac{\#\text{ Harmful}}{\#\text{ Harmless} + \#\text{ Harmful}}$$

$$\text{FOLLOWING RATE} = 1 - \frac{\#\text{ Irrelevant}}{\#\text{ All}}$$

Given a harmful prompt, we would expect the LLM to detect its malicious intent and refuse to engage. In the ideal case, we expect a safe LLM to have high FOLLOWING RATE but low HARMFUL RATE for each language.

Type	Language	Harmful (↓)	Following (↑)
High	Chinese	0	100
	Ruassian	2	99
	Spanish	0	100
	Portuguese	1	100
	French	0	100
	German	1	100
	Italian	1	100
	Dutch	1	99
	Turkish	1	98
Low	Hausa	32	76
	Armenian	26	82
	Igbo	38	72
	Javanese	34	79
	Kamba	28	65
	Halh	25	72
	Luo	28	75
	Maori	32	74
	Urdu	27	72

Table 1: A comparison of GPT-4's harmful and helpful rates in high- vs. low-languages. We observe that low-resource languages have a much higher harmful rate than high-resource ones, and low-resource languages achieve a much lower following rate than high-resource ones. ↓ means the lower the better, while ↑ means the opposite.

2.4 Two curses with low-resource languages

Curse of Harmful Response: Lower-resource languages lead to higher harmful rate. We show the HARMFUL RATE comparison between high- vs. low-resource languages in Table 1 Overall, we can see low-resource languages exhibit much higher HARMFUL RATE. The primary reason for this susceptibility might be the limited data available for alignment and pre-training, often leading to model jailbreaking. Consequently, LLM might produce harmful responses. This highlights the importance of dedicated resources toward model alignment and pre-training for these low-resource languages, ensuring inclusivity and reducing potential harm in LLM-driven applications.